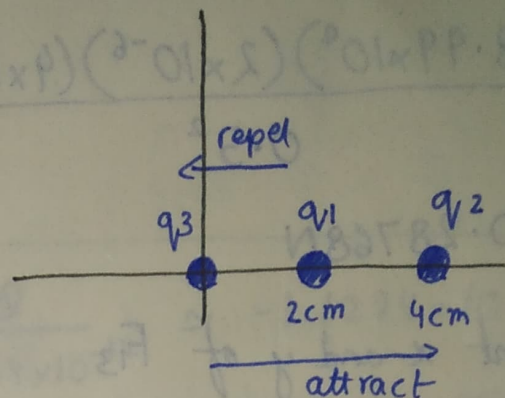


Applied Physics Assignment 3

Q1) $F_A = \frac{kq_1q_3}{r^2}$

$$F_A = \frac{(8.99 \times 10^9)(3 \times 10^{-9})(5 \times 10^{-9})}{(2 \times 10^{-2})^2}$$

$$F_A = 3.37 \times 10^{-4} \text{ N}$$



$$F_B = \frac{kq_2q_3}{r^2}$$

$$F_B = \frac{(8.99 \times 10^9)(7 \times 10^{-9})(5 \times 10^{-9})}{(4 \times 10^{-2})^2}$$

$$F_B = 1.96 \times 10^{-4} \text{ N}$$

$$F_{NET} = F_A - F_B$$

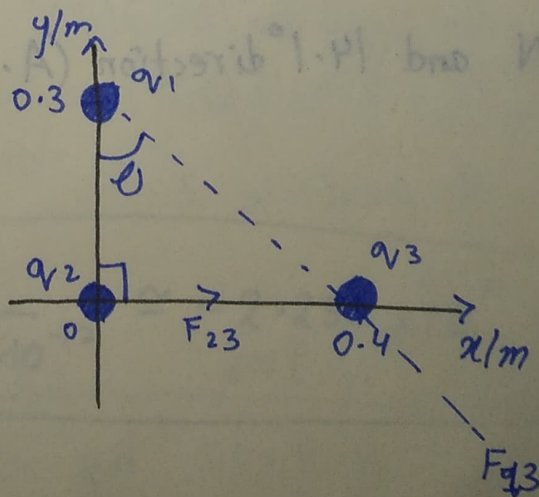
$$= (3.37 \times 10^{-4}) - (1.96 \times 10^{-4})$$

$$= 1.4 \times 10^{-4} \text{ N to the -ve x direction} \quad \text{Ans}$$

Q2) $F_{23} = \frac{kq_2q_3}{r^2}$

$$F_{23} = \frac{(8.99 \times 10^9)(2 \times 10^{-6})(4 \times 10^{-6})}{0.4^2}$$

$$F_{23} = 0.4495 \text{ N}$$



$$r = \sqrt{0.3^2 + 0.4^2} = 0.5 \text{ m}$$

$$\theta = \tan^{-1} \left(\frac{0.4}{0.3} \right) = 0.927 \text{ rad}$$

$$F_{13} = \frac{kq_1q_3}{r^2}$$

$$F_{13} = \frac{(8.99 \times 10^9)(2 \times 10^{-6})(4 \times 10^{-6})}{0.5^2}$$

$$F_{13} = 0.28768 \text{ N}$$

Component x and y of F_{13}

$$F_{13}(x) = 0.28768 \sin(0.927) = 0.230 \text{ N}$$

$$F_{13}(y) = 0.28768 \cos(0.927) = 0.172 \text{ N}$$

$$\begin{aligned} F_{\text{NET}}(x) &= F_{13}(x) + F_{23}(x) \\ &= 0.23 + 0.4495 \\ &= 0.6795 \text{ N} \end{aligned}$$

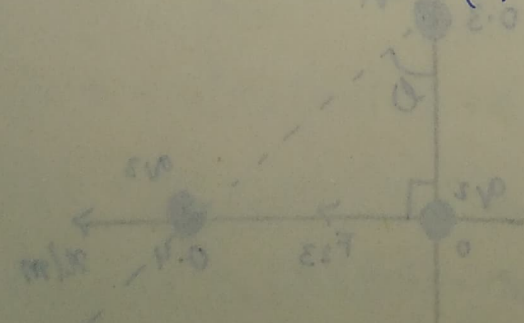
$$\begin{aligned} F_{\text{NET}}(y) &= F_{13}(y) + F_{23}(y) \\ &= 0.172 + 0 \\ &= 0.172 \text{ N} \end{aligned}$$

$$\begin{aligned} |F_{\text{NET}}| &= \sqrt{0.6795^2 + 0.172^2} \\ &= 0.700 \text{ N} \end{aligned}$$

$$\theta = \tan^{-1} \left(\frac{0.172}{0.6795} \right)$$

$$\theta = 0.247 \text{ rad or } 14.1^\circ$$

$\therefore 0.7 \text{ N}$ and 14.1° direction (A.C.W) from +ve x-axis Ans



Q3) for S_1

$$S_1 = -2Q + Q$$

$$S_1 = -Q$$

$$\phi = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$\phi = \frac{-Q}{\epsilon_0} \quad \text{or} \quad \frac{-Q}{8.54 \times 10^{-12}} \approx -1.1294 \times 10^{10} Q$$

for S_2

$$S_2 = +Q - Q$$

$$S_2 = 0$$

$$\phi = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$\phi = \frac{0}{\epsilon_0}$$

$$\phi = 0$$

for S_3

$$S_3 = -2Q + Q - Q$$

$$S_3 = -2Q$$

$$\phi = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$\phi' = \frac{-2Q}{\epsilon_0} \quad \text{or} \quad \frac{-2Q}{8.54 \times 10^{-12}} \approx -2.258 \times 10^{10} Q$$

for S_4

$$S_4 = 0 \quad \text{no charge}$$

$$\phi = \frac{q_{enc}}{\epsilon_0}$$

$$\phi = 0 \quad \text{Ans}$$

Q4) • charge of particle 1 = q_1

• charge of particle 2 = q_2

let $q_1 = q_2 = q$

• distance b/w charged particle = $3.2 \times 10^{-3} \text{ m}$

• particle 1 acceleration $a_1 = 7 \text{ m/s}^2$

• particle 2 acceleration $a_2 = 9 \text{ m/s}^2$

• Mass of particle 1 $m_1 = 6.3 \times 10^{-7} \text{ kg}$

• Mass of particle 2 $m_2 = ??$

for particle 1

$$F_1 = m_1 a_1$$

$$F_1 = (6.3 \times 10^{-7})(7)$$

$$F_1 = 4.41 \times 10^{-6} \text{ N}$$

a) Mass of particle 2

$$F_2 = m_2 a$$

$$4.41 \times 10^{-6} = 9 m_2$$

$$m_2 = 4.9 \times 10^{-7} \text{ kg} \quad \text{Ans}$$

$$b) F = \frac{kq_1q_2}{r^2}$$

$$4.41 \times 10^{-6} = \frac{(8.99 \times 10^9)(q^2)}{(3.2 \times 10^{-3})^2}$$

$$q = 7.09 \times 10^{-11} \text{ C ans}$$

$$Q5) F = \frac{kq_1q_2}{r^2}$$

$$1 = \frac{(8.99 \times 10^9)(q)(5 \times 10^5 - q)}{2^2}$$

$$4 = 4.495 \times 10^{15}q - 8.99 \times 10^9 q^2$$

$$(8.99 \times 10^9)q^2 - (4.495 \times 10^{15})q + 4 = 0$$

$$q = 5 \times 10^5 \text{ C or } q = 8.89 \times 10^{-16} \text{ C}$$

$\therefore$ charge on smallest sphere is $8.89 \times 10^{-16} \text{ C}$

Sir in Q5 I am confused whether charge should be $5 \times 10^{+5}$ or 5×10^{-5} . As the answer from +5 is quite odd and when I googled the question. Each had -5 as power. Therefore I am solving for both power to be on the safe side.

$$F = \frac{kq_1q_2}{r^2}$$

$$1 = \frac{(8.99 \times 10^9)(q)(5 \times 10^{-5} - q)}{2^2}$$

$$(8.99 \times 10^9)q^2 - (4.495 \times 10^5)q + 4 = 0$$

$$q = 3.84 \times 10^{-5} \text{ C or } q = 1.15 \times 10^{-5} \text{ C}$$

$\therefore$ charge on small sphere is $1.15 \times 10^{-5} \text{ C}$.